3	**(a)**	The gravitational potential <u>at infinity is set at zero</u> (as there is negligible g-force). Either of the following points, • The gravitational force is **attractive** and so the external force acting on the object is <u>opposite in direction</u> to the object's displacement as it moves from infinity to the point in question. • The celestial bodies are in a bound system. Work needs be given to separate them. Therefore, the system has negative potential. • At points that are less than infinity, the gravitation potential is negative as the test mass would lose gravitational potential energy when displaced from infinity (where GP is zero) to those points. Hence, the work done by the external force is negative which results in the gravitational potential having a negative value. <u>Alternative</u> The gravitational potential at infinity is set at zero. In bringing a point mass from infinity to a point in the gravitational field, the work done per unit mass by an external agent is **negative** as the force exerted on the point mass is opposite to its displacement from infinity. Hence, the gravitational potential has a negative value.	**B1** **B1** **A0**
	(b) (i)	The (net) gravitational field strength is equal to the negative gradient of the (net) gravitational potential – distance graph. Hence the direction of the net gravitational force is initially towards star A at $x = 0.1 \times 10^{12}$ m , zero at 0.52×10^{12} m and finally towards star B thereafter.	**B1** **B1**
	(b)(ii)	For the rock to reach the point where $x = 1.2 \times 10^{12}$ m, it must first be provided with a minimum amount of <u>energy to reach the point of maximum gravitational potential</u> (-4.4×10^8 J kg^{-1}). After that, the rock will be pulled towards star B. $GPE_A + KE_A = GPE_{max}$ (KE=0 at the peak of graph) $m\,\phi_A + \text{min KE} = m\,\phi_{max}$ $(180)(-10 \times 10^8) + (\tfrac{1}{2} \times 180 \times v^2) = (180)(-4.4 \times 10^8)$ $v = 3.3 \times 10^4$ m s^{-1}	 **M1** **A1**

4	**(a)**	amplitude to amplitude = 2 x_0 = 9.8 cm	